

MSE of Exponential Rate Estimators under Right Censoring: Simulation Study

2AMS11 Survival Analysis for Data Science

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Setup

Event times $T_i \stackrel{iid}{\sim} \text{Exp}(\lambda)$ with $\lambda = 0.05$, censoring times $C_i \stackrel{iid}{\sim} \text{Unif}(0, c_{\max})$ independent of T_i . We observe $Y_i = \min(T_i, C_i)$ and $\delta_i = \mathbb{I}\{T_i \leq C_i\}$, and compare

$$\hat{\lambda}_{\text{naive}} = \frac{1}{\bar{Y}}, \quad \hat{\lambda}_{\text{mle}} = \frac{r}{\sum_{i=1}^n Y_i}, \quad r = \sum_i \delta_i.$$

For each target censoring proportion $p_c \in \{0\%, 10\%, 50\%\}$ and sample size $n \in \{200, 500\}$ we run $R = 2000$ Monte Carlo replicates (seed 123). Since $P(T > C) = \frac{1 - e^{-\lambda c_{\max}}}{\lambda c_{\max}}$ does not depend on n , $c_{\max}$ is calibrated once per p_c on a pilot sample of 1000 event times by searching a grid of $c_{\max}$ values for the smallest deviation between the empirical censoring proportion and p_c , $p_c = 0\%$ is obtained by setting $C_i = \infty$. The search returns $c_{\max} = 186.5$ and 29.7 . Since $P(T > C)$ is very flat in $c_{\max}$ at small p_c and each grid point uses one pilot draw, the realised proportions (10.7%, 52.1%) sit slightly above target, so we report realised values throughout. Per replicate we record $\hat{\lambda}_{\text{naive}}$, $\hat{\lambda}_{\text{mle}}$ (recorded as NA when $r = 0$) and the realised censoring fraction. Bias, variance and $\text{MSE} = \widehat{\text{bias}}^2 + \widehat{\text{var}}$ are the Monte Carlo averages over the non-NA replicates.

Results

Monte Carlo results over $R = 2000$ replicates, $\lambda = 0.05$, seed 123 and "obs cens" is the realised censoring fraction averaged over replicates.

n	p_c	obs cens	$\hat{\lambda}_{\text{naive}}$				$\hat{\lambda}_{\text{mle}}$				$P(r = 0)$
			mean	bias	var	MSE	mean	bias	var	MSE	
500	0%	0.000	0.05014	0.00014	5.27e-6	5.29e-6	0.05014	0.00014	5.27e-6	5.29e-6	0
500	10%	0.107	0.05613	0.00613	6.15e-6	4.38e-5	0.05013	0.00013	5.51e-6	5.52e-6	0
500	50%	0.521	0.10469	0.05469	1.34e-5	3.00e-3	0.05018	0.00018	1.01e-5	1.01e-5	0

Discussion

No censoring. With $p_c = 0$ all $\delta_i = 1$, so $r = n$ and the two estimators coincide exactly. Both show the small positive $O(1/n)$ bias of $n/\sum_i T_i$ and an MSE of order λ^2/n .

Effect of censoring on the naive estimator. Censoring replaces T_i by the smaller value $Y_i = \min(T_i, C_i)$, so $\mathbb{E}[\bar{Y}] < 1/\lambda$ and $\hat{\lambda}_{\text{naive}} = 1/\bar{Y}$ is biased upward, meaning the estimator behaves as if subjects fail at the moment they are lost to follow-up, which makes the event rate look larger than it is. The bias grows steeply with p_c (+0.006 at $p_c \approx 11\%$, +0.055 at $p_c \approx 52\%$, where the estimate roughly doubles). Indeed, the exponential has constant hazard λ , so its density satisfies $f_T = \lambda S_T$. Integrating this against the censoring survival function gives $\mathbb{E}[\delta] = \lambda \mathbb{E}[Y]$. This shows that the rate equals expected events divided by expected time at risk. Since $\mathbb{E}[\delta] = 1 - p_c$, this yields $\mathbb{E}[Y] = (1 - p_c)/\lambda$, meaning censoring shortens mean follow-up by exactly the censored fraction. The naive estimator discards the event count and therefore converges to $\lambda/(1 - p_c)$, which is an asymptotic bias of $\lambda p_c/(1 - p_c)$, which well approximates both values. Its variance barely changes. Because the bias enters the MSE squared and is $O(1)$ in n , it dominates as soon as censoring is non-negligible: at $p_c \approx 52\%$ the naive MSE is 110× the MLE's, and it does not improve when n grows from 200 to 500 which means the estimator doesn't converge under censoring.

The MLE. $\hat{\lambda}_{\text{mle}}$ divides the number of observed events by the total time at risk, which is exactly the sufficient statistic of the censored exponential likelihood $L(\lambda) = \lambda^r e^{-\lambda \sum_i y_i}$. Its bias stays negligible at every censoring level, and its variance grows only mildly, approximately $\lambda^2/\mathbb{E}[r] = \lambda^2/(n(1 - p_c))$. Therefore censoring costs information but introduces no systematic error. The MSE therefore roughly doubles from $p_c = 0\%$ to 52% , and falls when n goes from 200 to 500, as expected for a $\sqrt{n}$ -consistent estimator.

Use of AI tools

- **Date of use:** 16 September
 - **Tool:** Claude Opus 5
 - **Exact prompt(s):** "Explain step 2 from the simulation procedure regarding the part with $c_{\max}$."; "Have I missed anything in my code?"
 - **Summary of output:** For the first prompt, the explanation given said that without calibration, just generating a uniform distribution for censoring will not allow us to have any control over the percentages, smaller $c_{\max}$ risks censoring too much and larger $c_{\max}$ risks censoring too little. The solution offered was to try multiple values. The second prompt only pointed out a few naming issues for some functions.
 - **How it was used:** The first prompt was used so that we could better understand the assignment and choose a proper $c_{\max}$ value and the second prompt was meant to verify whether the code fits the assignment.
 - **Modifications:** Modifications were made to some functions that were using global values instead of the given parameters.
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- **Date of use:** 17 September
 - **Tool:** Claude Opus 5
 - **Exact prompt(s):** "Give me, in LaTeX only, the mathematical formulas behind Assignment1Code.R: exponential event times, uniform censoring times, $Y = \min(T, C)$, the event indicator, the censoring probability and the grid calibration of $c_{\max}$, the censored likelihood with both estimators derived from it, the Monte Carlo definitions of bias, variance and MSE, and the asymptotic results for the naive estimator and the MLE. Use amsmath."
 - **Summary of output:** The output was comprised of the exact formulas in LaTeX.
 - **How it was used:** We used the given formulas in order to append them directly in the report as some of them were hard to write manually in LaTeX.
 - **Modifications:** The generated formulas were pasted where they were needed inside the report.

Contribution statement

- **Avram Andrei:** Wrote the report.
- **Mihaiescu Cristian:** Contributed in the brainstorming, will have more responsibilities in next assignments.
- **Szelestey Adam:** Proofread and revised the final deliverables.
- **Vrincianu Darius:** Created the repository and wrote the code.
- **Zeng Qin Tao:** Contributed in the brainstorming, will have more responsibilities in next assignment.